Multi-Source Candidate Data Transformer

Design One-Pager - Eightfold Engineering Intern Assignment

1. The Problem

Eightfold ingests candidate information from structured sources (Recruiter CSV, ATS JSON, GitHub API) and unstructured sources (Resumes, Recruiter Notes). Downstream systems require one clean, canonical profile per candidate with normalized formats, deduplicated entries, and detailed provenance indicating source reliability.

2. Pipeline Architecture

The deterministic, rule-based pipeline consists of six stages:

1. Detect & Extract: Source-specific extractors parse inputs into unified CandidateFragment objects.
2. Normalize: Phones map to E.164, locations extract city/country, degrees map to standard formats.
3. Conflict Analysis (CACS): Analyzes divergence between overlapping fields. Flags silent conflict masking, creates a traceable Lineage IR log, and applies temporal stratification to skills.
4. Merge: Identities match via primary (email) and secondary (name + phone) keys. Scalar fields take the most reliable source's value. Array fields (Skills, Experience, Education) merge via fuzzy matching deduplication.
5. Confidence Scoring: Each profile receives an overall confidence score based on source weights (ATS > CSV > Resume), cross-source agreement, and penalties from detected conflicts.
6. Project: Runtime JSON configurations dictate field selection, JSON structure mapping, and missing-value policies (null vs omit vs error).
7. Validate: The final payload conforms rigidly to the requested downstream schema.

3. Resolution Heuristics

- Identity Merging: Uses a multi-pass approach matching overlapping emails, then phone/name heuristics.
- Experience/Education Dedup: Employs rapidfuzz matching (threshold > 75%) on institution and company names, along with date range intersections.
- Unstructured Parsing: Regex-driven contextual extraction handles phrases like 'works as a PM at Flipkart' to properly anchor both title and company, minimizing hallucinations.

4. Future Enhancements (Descoped for MVP)

- ML-based sequence tagging (NER) for free-text sections, complementing regex patterns.
- Parallelized I/O streaming for processing gigabyte-scale exports.
- Direct database sync/upsert capabilities beyond JSON file generation.